

SKA Scanning Strategy: To-do list

Models

- ☐ Foregrounds:
 - ☐ Include Phil and Zheng's $C_l(\nu, \nu')$ model
 - ☐ Include Mel's CNN/scaled Haslam model for galactic emission
- ☐ Include an alt-elevation-dependent ground spill model
- ☐ Include an elevation-dependent atmosphere model
- ☐ Beam:
 - ☐ Simple Gaussian beam
 - ☐ Gaussian beam with ~ 1 degree FWHM ripples in the width

Notes:

Simulations

- ☐ Include the 50 MHz band at $z = 0.8$, with ~ 250 kHz channels
- ☐ Test different cases of $1/f$ noise, e.g. $f_k \sim 1/\text{seconds}$ to $\sim 1/\text{tens of minutes}$
- ☐ Scan strategies:
 - ☐ Maximum speed, constant elevation scan
 - ☐ Medium speed, constant elevation scan
 - ☐ Drift scan:
 - ☐ Fixed alt/az scans
 - ☐ 1x-2x MeerKLASS size patches
 - ☐ Including rising and setting

Notes: Different scanning strategies available in MuSEEK.

Maps

- ☐ For all of the above:
 - ☐ Simple filtering map-maker
 - ☐ Bayesian map-maker with moderate priors
 - ☐ Map-level comparisons

Notes: